

Операционные системы

Продвинутое использование git

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Получение навыков правильной работы с репозиториями git.

- 1 Выполнить работу для тестового репозитория.
- 2 Преобразовать рабочий репозиторий в репозиторий с git-flow и conventional commits.

2. Процесс выполнения лабораторной работы

```
mrfilosophova@mrfilosophova:~$ pnpm setup
Appended new lines to /home/mrfilosophova/.bashrc

Next configuration changes were made:
export PNPM_HOME="/home/mrfilosophova/.local/share/pnpm"
case ":$PATH:" in
  *":$PNPM_HOME:") ;;
  *) export PATH="$PNPM_HOME:$PATH" ;;
esac

To start using pnpm, run:
source /home/mrfilosophova/.bashrc
mrfilosophova@mrfilosophova:~$ source ~/.bashrc
mrfilosophova@mrfilosophova:~$
```

Рисунок 1: Node.js

```
mrfilosophova@mrfilosophova:~$ pnpm add -g commitizen
```

```
Update available! 10.27.0 → 11.25.0.  
Changelog: https://pnpm.io/v/11.25.0  
To update, run: pnpm self-update
```

```
  
WARN GET https://registry.npmjs.org/fs-extra error (ECONNRESET). Will retry in 10 seconds.  
2 retries left.  
WARN Request took 31590ms: https://registry.npmjs.org/fs-extra  
WARN GET https://registry.npmjs.org/ora error (ECONNRESET). Will retry in 10 seconds. 2 re  
tries left.  
WARN GET https://registry.npmjs.org/through error (ECONNRESET). Will retry in 10 seconds.  
2 retries left.  
WARN GET https://registry.npmjs.org/merge error (ECONNRESET). Will retry in 10 seconds. 2
```

Рисунок 2: установка commitizen


```
mrfilosophova@mrfilosophova:~$ pnpm add -g standard-changelog
WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +14
++++
Progress: resolved 181, reused 148, downloaded 14, added 14, done

/home/mrfilosophova/.local/share/pnpm/global/5:
+ standard-changelog 8.1.0

Done in 1.6s using pnpm v10.27.0
mrfilosophova@mrfilosophova:~$
```

Рисунок 3: установка standard-changelog

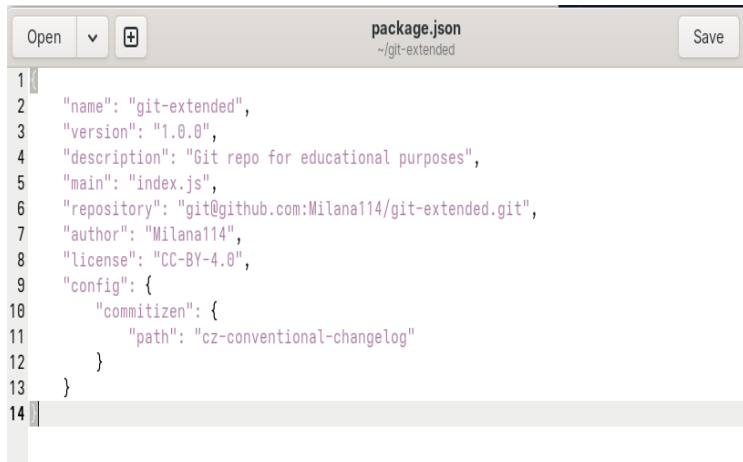


Рисунок 4: package.json

Использование commitizen

```
mrfilosophova@mrfilosophova:~/git-extended$ git add .
mrfilosophova@mrfilosophova:~/git-extended$ git cz
cz-cli@4.3.2, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:    Other changes that don't
modify src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
  (3) yes
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master 511649a] chore: yes
 1 file changed, 14 insertions(+)
 create mode 100644 package.json
mrfilosophova@mrfilosophova:~/git-extended$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 478 bytes | 478.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Milana114/git-extended.git
 84d0281..511649a master -> master
mrfilosophova@mrfilosophova:~/git-extended$
```

Рисунок 5: Отправка

```
mrfilosophova@mrfilosophova:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? []
Hooks and filters directory? [/home/mrfilosophova/git-extended/.git/hooks]
mrfilosophova@mrfilosophova:~/git-extended$ git branch
* develop
  master
mrfilosophova@mrfilosophova:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:   https://github.com/Milana114/git-extended/pull/new/develop
remote:
To github.com:Milana114/git-extended.git
* [new branch]      develop -> develop
mrfilosophova@mrfilosophova:~/git-extended$
```

Рисунок 6: Инициализация и загрузка изменений

```
mrfilosophova@mrfilosophova:~/git-extended$ standard-changelog --first-release
mrfilosophova@mrfilosophova:~/git-extended$ git add CHANGELOG.md
mrfilosophova@mrfilosophova:~/git-extended$ git commit -am 'chore(site): add changelog'
[release/1.0.0 2707b23] chore(site): add changelog
 1 file changed, 1 insertion(+)
 create mode 100644 CHANGELOG.md
mrfilosophova@mrfilosophova:~/git-extended$ git flow release finish 1.0.0
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
Merge made by the 'ort' strategy.
  CHANGELOG.md | 1 +
  1 file changed, 1 insertion(+)
 create mode 100644 CHANGELOG.md
Already on 'master'
Your branch is ahead of 'origin/master' by 2 commits.
  (use "git push" to publish your local commits)
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Merge made by the 'ort' strategy.
  CHANGELOG.md | 1 +
  1 file changed, 1 insertion(+)
 create mode 100644 CHANGELOG.md
Deleted branch release/1.0.0 (was 2707b23).

Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'master'
- The release was tagged '1.0.0'
- Release tag '1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
```

Рисунок 7: Завершение релиза

Отправим данные на github

```
mrfilosophova@mrfilosophova:~/git-extended$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 544 bytes | 544.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To github.com:Milana114/git-extended.git
    511649a..d613f11  develop -> develop
    511649a..1f2c9fb  master -> master
mrfilosophova@mrfilosophova:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 155 bytes | 155.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Milana114/git-extended.git
 * [new tag]         1.0.0 -> 1.0.0
mrfilosophova@mrfilosophova:~/git-extended$ gh release create v1.0.0 -F CHANGELOG.md
https://github.com/Milana114/git-extended/releases/tag/v1.0.0
mrfilosophova@mrfilosophova:~/git-extended$
```

Рисунок 8: Отправка

```
mrfilosophova@mrfilosophova:~/git-extended$ git flow feature start feature_branch
Switched to a new branch 'feature/feature_branch'

Summary of actions:
- A new branch 'feature/feature_branch' was created, based on 'develop'
- You are now on branch 'feature/feature_branch'

Now, start committing on your feature. When done, use:

    git flow feature finish feature_branch

mrfilosophova@mrfilosophova:~/git-extended$ touch test2
mrfilosophova@mrfilosophova:~/git-extended$ git flow feature finish feature_branch
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Already up to date.
Deleted branch feature/feature_branch (was d613f11).

Summary of actions:
- The feature branch 'feature/feature_branch' was merged into 'develop'
- Feature branch 'feature/feature_branch' has been locally deleted
- You are now on branch 'develop'

mrfilosophova@mrfilosophova:~/git-extended$
```

Рисунок 9: Объединение веток

```
Summary of actions:
- Release branch 'release/1.2.3' has been merged into 'master'
- The release was tagged '1.2.3'
- Release tag '1.2.3' has been back-merged into 'develop'
- Release branch 'release/1.2.3' has been locally deleted
- You are now on branch 'develop'

mrfilosophova@mrfilosophova:~/git-extended$ git push --all
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 746 bytes | 746.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:Milana114/git-extended.git
   d613f11..54249a4  develop -> develop
   1f2c9fb..838f440  master -> master
mrfilosophova@mrfilosophova:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 156 bytes | 156.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Milana114/git-extended.git
 * [new tag]         1.2.3 -> 1.2.3
mrfilosophova@mrfilosophova:~/git-extended$ gh release create v1.2.3 -F CHANGELOG.md
https://github.com/Milana114/git-extended/releases/tag/v1.2.3
mrfilosophova@mrfilosophova:~/git-extended$
```

Рисунок 10: Завершение релиза

Подготовка рабочего репозитория

```
mrfilosophova@mrfilosophova:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git cz
cz-cli@4.3.2, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:   Other changes that don't
modify src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
  (4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master 04c015c] chore: test
 1 file changed, 11 insertions(+), 11 deletions(-)
mrfilosophova@mrfilosophova:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 475 bytes | 475.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Milana114/2026-1--study--os-intro.git
   ec4d959..04c015c  master -> master
mrfilosophova@mrfilosophova:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
```

Рисунок 11: package.json и коммит

```
Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'master'
- The release was tagged '1.0.0'
- Release tag '1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'

mrfilosophova@mrfilosophova:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 634 bytes | 634.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To github.com:Milana114/2026-1--study--os-intro.git
   04c015c..87a19d1  develop -> develop
   04c015c..7c491ba  master -> master
mrfilosophova@mrfilosophova:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 156 bytes | 156.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Milana114/2026-1--study--os-intro.git
 * [new tag]           1.0.0 -> 1.0.0
mrfilosophova@mrfilosophova:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
```

Рисунок 12: Завершение релиза

3. Выводы по проделанной работе

Мы приобрели практические навыки взаимодействия с дополнительными функциями гитхаб.